

Precalc Lesson 0.2 Homework

Answer Key

1. Center = $(7, -2)$, Radius = 10

2. $(x - 6)^2 + y^2 = 9$

3. Center = $(-13, -8)$, Radius = 14

4. B

5. $(x - 5)^2 + (y + 1)^2 = 169$

6. $(x - 4)^2 + (y - 1)^2 = 196$

7. B

8. $(x + 8)^2 + (y + 5)^2 = 292$

9. The equation represents a circle centered at $(2, 9)$ with a radius of 12 . The distance between $(7, -2)$ and $(2, 9)$ is $\sqrt{146}$ which is greater than $\sqrt{144} = 12$. Thus $(7, -2)$ is outside the circle.

10. a. $r_1 = 8$, $r_2 = 1$

Multiple solutions are possible as long as $5 + r_1 > 9 + r_2$

b. $r_1 = 3, r_2 = 4$

Multiple solutions are possible as long as $r_1 + r_2 > 4$ and $|r_1 - r_2| < 4$.

c. $(x - 5)^2 + (y - 7)^2 = 4$ and $(x - 9)^2 + (y - 7)^2 = 4$

Multiple solutions are possible as long as $r_1 + r_2 = 4$ or $|r_1 - r_2| = 4$.